Understanding Flow Matching from a mathematical view

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October 30, 2025

Outline

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Probability Path and Coupling

- Let $\{p_t\}_{t\in[0,1]}$ be a prescribed path of marginals on $\mathbb{R}^d$ with p_0 simple and p_1 the data.
- A *coupling* induces random trajectories $t \mapsto x_t$ with instantaneous velocity $\dot{x}_t = \frac{\mathrm{d}}{\mathrm{d}t}x_t$ (may be stochastic given x_t).
- Goal: learn a *deterministic* vector field $v : \mathbb{R}^d \times [0,1] \to \mathbb{R}^d$ such that the ODE

$$\frac{\mathrm{d}}{\mathrm{d}t}x_t = v(x_t, t), \qquad x_0 \sim p_0,$$

yields marginals $x_t \sim p_t$ for all t.

Strong and Weak Continuity

From the Fokker–Planck equation, for $\frac{\mathrm{d}x}{\mathrm{d}t} = v(x,t)$, we have

$$\partial_t p_t(x) + \nabla \cdot \big(p_t(x)\, v(x,t)\big) = 0.$$

The corresponding weak form of above strong form is

Weak form (for any test functions $\varphi \in C_c^\infty(\mathbb{R}^d)$)

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{\mathbb{R}^d} \varphi(x)\, p_t(x)\, \mathrm{d}x \;=\; \int_{\mathbb{R}^d} \nabla\varphi(x) \cdot v(x,t)\, p_t(x)\, \mathrm{d}x,$$

i.e.,

$$\frac{\mathrm{d}}{\mathrm{d}t} \mathbb{E}[\varphi(x_t)] = \mathbb{E}[\nabla\varphi(x_t) \cdot v(x,t)].$$

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Weak Form from Random Trajectories

Let x_t be the coupled trajectory (not necessarily deterministic). For any test function $\phi \in C_c^\infty(\mathbb{R}^d)$, we have

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\mathbb{E}[\varphi(x_t)] = \mathbb{E}[\nabla\varphi(x_t)\cdot\dot{x}_t],$$

by the expectation of the derivative of the test function.
Expanding the expectation into integrals at a fixed t:

$$\begin{aligned}
\mathbb{E}[\nabla\varphi(x_t)\cdot\dot{x}_t] &= \iint_{\mathbb{R}^d\times\mathbb{R}^d} \nabla\varphi(x)\cdot\dot{x}\;\mu_t(\,\mathrm{d}x,\,\mathrm{d}\dot{x}) \\
&= \int_{\mathbb{R}^d} \nabla\varphi(x)\cdot\Big(\int \dot{x}\,p(\dot{x}\,|\,x,t)\,\mathrm{d}\dot{x}\Big)\,p_t(x)\,\mathrm{d}x \\
&= \int_{\mathbb{R}^d} \nabla\varphi(x)\cdot v^\star(x,t)\,p_t(x)\,\mathrm{d}x = \mathbb{E}[\nabla\varphi(x_t)\cdot v^\star(x,t)].
\end{aligned}$$

where $v^\star(x,t) \equiv \mathbb{E}[\dot{x}_t \mid x_t = x,\ t] = \int_{\mathbb{R}^d} \dot{x}\,p(\dot{x}\,|\,x,t)\,\mathrm{d}\dot{x}$. Thus the random path induces the weak identity with $v = v^\star$.

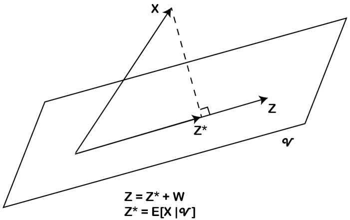
Theoretical Support: Conditional Expectation

Theorem

Let X be an integrable random variable. Then for each σ-algebra $\mathcal{V}$ and $Z \in \mathcal{V}$, $Z^* = \mathbb{E}(X|\mathcal{V})$ solves the least square problem

$$argmin_{Z \in \mathcal{V}} \|Z - X\| \,,$$

where $\|Z\| = \left(\int Z^2 dP\right)^{\frac{1}{2}}$.



Hence we have $L = \mathbb{E}\|v_\theta(x_t, t) - \dot{x}_t\|^2$.

Practical Training Recipe (Per-Iteration)

1. Sample $t \sim w(t)$ (typically uniform on $[0, 1]$).
2. Sample $(x_t, \dot{x}_t)$ from the chosen path coupling.
3. Regress $\dot{x}_t$ on (x_t, t) via the MSE:

$$\mathcal{L} = \left\| v_\theta(x_t, t) - \dot{x}_t \right\|^2.$$

4. Update θ by SGD; no density estimation, no reverse simulation is needed during training.

Remark

Different path/coupling choices change $v^\star$ and the irreducible variance $\mathbb{E}\|\dot{x}_t - v^\star\|^2$, affecting sample efficiency and optimization, but the optimal target remains the conditional mean.

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Key Takeaways

- The correct notion of "matching the path" is the **weak continuity** identity for all test functions.
- The only deterministic field depending on (x, t) that preserves this identity is the **conditional mean velocity**

$$\boxed{v^\star(x,t) = \mathbb{E}[\dot{x}_t \mid x_t = x,\ t]}.$$

- The **flow-matching MSE** is precisely the L^2 projection selecting $v^\star$:

$$v_\theta \xrightarrow[\text{minimize}]{\mathbb{E}\|v_\theta(x_t,t)-\dot{x}_t\|^2} v^\star.$$

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Fokker–Planck / Forward Kolmogorov

Theorem (Fokker–Planck / Forward Kolmogorov)

Let $(X_t)_{t\geq 0} \in \mathbb{R}^N$ solve the Itô SDE

$$dX_t = \mu(X_t, t)\,dt + \boldsymbol{\sigma}(X_t, t)\,dW_t,$$

where $\mu : \mathbb{R}^N \times [0,\infty) \to \mathbb{R}^N$ is the drift, $\boldsymbol{\sigma} : \mathbb{R}^N \times [0,\infty) \to \mathbb{R}^{N\times M}$ the diffusion, and W_t an M-dimensional standard Wiener process. Define the diffusion tensor

$$D(x,t) := \frac{1}{2}\boldsymbol{\sigma}(x,t)\,\boldsymbol{\sigma}(x,t)^\top \in \mathbb{R}^{N\times N}.$$

Then the (smooth) density $p(x,t)$ satisfies the Fokker–Planck equation

$$\boxed{\partial_t p(x,t) = -\sum_{i=1}^{N} \frac{\partial}{\partial x_i}\big[\mu_i(x,t)\,p(x,t)\big] + \sum_{i,j=1}^{N} \frac{\partial^2}{\partial x_i\,\partial x_j}\big[D_{ij}(x,t)\,p(x,t)\big]}$$

Isotropic diffusion: $\boldsymbol{\sigma}(x,t) = \sigma(t)\,\mathbf{I}_N$

When $\boldsymbol{\sigma}(x,t) = \sigma(t)\,\mathbf{I}_N$, the diffusion tensor becomes

$$D(x,t) = \tfrac{1}{2}\,\sigma(t)^2\,\mathbf{I}_N, \qquad D_{ij}(x,t) = \tfrac{1}{2}\,\sigma(t)^2\,\delta_{ij}.$$

Hence the Fokker–Planck equation simplifies to the drift–diffusion form

$$\boxed{\partial_t p(x,t) = -\nabla\cdot\big(\mu(x,t)\,p(x,t)\big) \;+\; \frac{\sigma(t)^2}{2}\,\Delta p(x,t)}$$

where $\nabla\cdot(\mu p) = \sum_{i=1}^N \partial_{x_i}\big(\mu_i p\big)$ and $\Delta = \sum_{i=1}^N \partial_{x_i}^2$ is the Laplacian.

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Mathematical connection to diffusion models

Theoretical link: For a forward SDE $\mathrm{d}\mathbf{x} = f(\mathbf{x}, t)\,\mathrm{d}t + g(t)\,\mathrm{d}B_t$, the probability flow ODE is:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x} = f(\mathbf{x}, t) - \tfrac{1}{2}g^2(t)\,\nabla_{\mathbf{x}} \log p(\mathbf{x}, t). \tag{1}$$

Key mathematical insight: When Flow Matching uses Gaussian paths that match the SDE's marginal distributions p_t, the learned velocity field $\mathbf{v}_\theta$ approximates the probability flow drift:

$$\mathbf{v}_\theta(\mathbf{x}, t) \approx f(\mathbf{x}, t) - \tfrac{1}{2}g^2(t)\,\nabla_{\mathbf{x}} \log p(\mathbf{x}, t) \tag{2}$$

Crucial advantage: Flow Matching achieves this approximation *without* ever computing the score $\nabla_{\mathbf{x}} \log p(\mathbf{x}, t)$ during training!
Practical implications: This connection allows Flow Matching to inherit the theoretical guarantees of diffusion models while avoiding their computational bottlenecks.

Mathematical derivation: probability flow ODE

Setup: Forward SDE $\mathrm{d}\mathbf{x} = f(\mathbf{x}, t)\,\mathrm{d}t + g(t)\,\mathrm{d}B_t$ with density $p(\mathbf{x}, t)$ solving the Fokker-Planck equation:

$$\partial_t p = -\nabla \cdot (fp) + \frac{1}{2} g^2 \triangle p. \tag{3}$$

Key insight: A deterministic ODE $\dot{\mathbf{x}} = \mathbf{v}(\mathbf{x}, t)$ transports densities via the continuity equation:

$$\partial_t p + \nabla \cdot (p\,\mathbf{v}) = 0. \tag{4}$$

Mathematical derivation: Equating the two equations:

$$-\nabla \cdot (p\,\mathbf{v}) = -\nabla \cdot (fp) + \frac{1}{2} g^2 \triangle p \tag{5}$$

$$\Rightarrow\ \mathbf{v} = f - \frac{1}{2} g^2 \frac{\nabla p}{p}\ =\ f - \frac{1}{2} g^2 \nabla \log p. \tag{6}$$

Result: The probability flow ODE drift is $\mathbf{v} = f - \frac{1}{2} g^2 \nabla \log p$.
Connection to Flow Matching: When Flow Matching learns $\mathbf{v}_\theta$, it approximates this probability flow drift without computing the score $\nabla \log p$.

Questions

From

1. Score model $s_\theta(x_t) \to s_{\theta^*}(x_t) \equiv \mathbb{E}[\nabla \log p(x_t|x_0) \mid x_t]$
2. X prediction model (denoising model, $x0$ model) $x_\theta(x_t) \to x_{\theta^*}(x_t) \equiv \mathbb{E}[x_0 \mid x_t]$;
3. ϵ prediction model $\epsilon_\theta(x_t) \to \epsilon_{\theta^*}(x_t) \equiv \mathbb{E}[\epsilon \mid x_t]$.

and

$$v^\star(x_t) = \mathbb{E}[\dot{x}_t \mid x_t]$$

, what is the relationship between optimal $v_\theta(x,t)$ and the above models? Hints:

$$v_{\theta^*}(x_t) = \mathbb{E}[\dot{x}_t \mid x_t] = \mathbb{E}[\dot{\alpha}(t)x_t + \dot{\sigma}(t)\epsilon \mid x_t]. \tag{7}$$

and

$$s(x_t) = \mathbb{E}_x[\nabla_{x_t} \log p(x_t|x_0)|x_t] = \mathbb{E}_x[-\frac{x_t - \alpha x_0}{\sigma^2}|x_t] = -\frac{x_t - \alpha\mathbb{E}[x_0|x_t]}{\sigma^2}. \tag{8}$$

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Gaussian conditional paths: mathematical derivation

Assumption: Gaussian conditional paths

$$q_t(\mathbf{x} \mid \mathbf{x}_0) = \mathcal{N}\left(\mathbf{x};\ \boldsymbol{\mu}_t(\mathbf{x}_0),\, \sigma_t^2 \mathbf{I}\right), \quad \mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_0) + \sigma_t \boldsymbol{\epsilon}. \tag{9}$$

Mathematical derivation: Taking the time derivative of the path:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x}_t = \dot{\boldsymbol{\mu}}_t(\mathbf{x}_0) + \dot{\sigma}_t\, \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} = \frac{\mathbf{x}_t - \boldsymbol{\mu}_t(\mathbf{x}_0)}{\sigma_t}. \tag{10}$$

Conditional expectation: Since $\boldsymbol{\epsilon}$ is independent of $\mathbf{x}_t$ given $\mathbf{x}_0$, we have:

$$\boxed{v = \dot{\boldsymbol{\mu}}_t(\mathbf{x}_0) + \frac{\dot{\sigma}_t}{\sigma_t}\left(\mathbf{x} - \boldsymbol{\mu}_t(\mathbf{x}_0)\right).} \tag{11}$$

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Mathematical analysis of path choices

1. **Linear bridge** (optimal transport):
   - $\mu_t(\mathbf{x}_0) = (1-t)\,\mathbf{x}_0$, $\sigma_t = t\,\sigma_{\max}$
   - **Mathematical property**: Minimizes Wasserstein-2 distance
   - **Velocity**: $v = \frac{\sigma_{\max}}{t}(\mathbf{x} - \mathbf{x}_0)$
   - **Advantage**: Direct connection to optimal transport theory
2. **VP-like** (DDPM-inspired):
   - $\mu_t(\mathbf{x}_0) = \alpha(t)\,\mathbf{x}_0$, $\sigma_t = \sqrt{1-\alpha^2(t)}$
   - **Mathematical property**: Preserves energy structure
   - **Velocity**: $v = \dot{\alpha}(t)\mathbf{x}_0 + \frac{\dot{\sigma}_t}{\sigma_t}(\mathbf{x} - \alpha(t)\mathbf{x}_0)$
   - **Advantage**: Compatible with existing diffusion model schedules
3. **VE-like** (SMLD-inspired):
   - $\mu_t(\mathbf{x}_0) = \mathbf{x}_0$, $\sigma_t = \sigma(t)$ increasing
   - **Mathematical property**: Preserves data structure
   - **Velocity**: $v = \frac{\dot{\sigma}(t)}{\sigma(t)}(\mathbf{x} - \mathbf{x}_0)$
   - **Advantage**: Simple form, good for high-dimensional data

Key insight: All choices provide *closed-form* $\mathbf{u}_t$ via the boxed formula!

Derivation: VE/VP target forms

VE-like. $\boldsymbol{\mu}_t(\mathbf{x}_0) = \mathbf{x}_0$, $\sigma_t = \sigma(t)$. Then

$$v = 0 + \frac{\dot{\sigma}_t}{\sigma_t}\,(\mathbf{x} - \mathbf{x}_0). \tag{12}$$

VP-like. $\boldsymbol{\mu}_t(\mathbf{x}_0) = \alpha(t)\,\mathbf{x}_0$, $\sigma_t = \sqrt{1-\alpha^2(t)}$. Using $\mathbf{u}_t = \dot{\boldsymbol{\mu}}_t + (\dot{\sigma}_t/\sigma_t)(\mathbf{x} - \boldsymbol{\mu}_t)$, we get

$$\begin{aligned} v &= \dot{\alpha}\,\mathbf{x}_0 + \frac{\dot{\sigma}_t}{\sigma_t}\,\mathbf{x} - \frac{\dot{\sigma}_t}{\sigma_t}\,\alpha\,\mathbf{x}_0 && (13)\\ &= \frac{\dot{\sigma}_t}{\sigma_t}\,\mathbf{x} + \left(\dot{\alpha} - \alpha\,\frac{\dot{\sigma}_t}{\sigma_t}\right)\mathbf{x}_0. && (14) \end{aligned}$$

No explicit form for $\dot{\sigma}_t/\sigma_t$ is required to train; schedules determine it numerically.
Mathematical insight: These derivations show how different path choices lead to different velocity field structures, all computable analytically.

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Flow Matching: mathematical training objective

Theoretical foundation: Given conditional paths $q_t(\mathbf{x} \mid \mathbf{x}_0)$, the optimal velocity field satisfies:

$$v^\star(x,t) = \mathbb{E}[\dot{x}_t \mid x_t = x,\ t] \tag{15}$$

Training procedure:

1. Sample $t \sim \mathcal{U}[0,1]$, $\mathbf{x}_0 \sim p_{\text{data}}$, $\mathbf{x}_t \sim q_t(\cdot \mid \mathbf{x}_0)$
2. Compute analytic target $\mathbf{u}_t(\mathbf{x}_t \mid \mathbf{x}_0)$ from chosen path
3. Minimize regression loss

Mathematical objective:

$$J_{\text{FM}} = \mathbb{E}\big[\,\|\,\mathbf{v}_\theta(\mathbf{x}_t,t) - v\,\|^2\,\big]. \tag{16}$$

Key mathematical advantage: No score estimation needed; all targets are analytic!

Computational benefits:

- Stable gradients (no exploding/vanishing)
- Parallelizable across time steps
- No special architectural requirements

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Mathematical foundation of inference

Theoretical basis: The learned velocity field $\mathbf{v}_\theta(\mathbf{x}, t)$ defines a deterministic ODE:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x} = \mathbf{v}_\theta(\mathbf{x}, t), \quad t \in [0, 1] \tag{17}$$

Mathematical properties:

- **Existence**: Under Lipschitz conditions, solutions exist and are unique
- **Consistency**: The ODE preserves the continuity equation
- **Stability**: Deterministic integration avoids accumulation of stochastic errors

Generation process:

1. Initialize $\mathbf{x}(1) \sim p_1$ (e.g., $\mathcal{N}(0, \mathbf{I})$)
2. Integrate ODE backwards: $\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x} = \mathbf{v}_\theta(\mathbf{x}, t)$, $t : 1 \to 0$
3. Use standard ODE solvers (Euler, RK4, Dormand-Prince, etc.)

Mathematical advantages:

- Deterministic sampling (reproducible)
- Fast with large solver steps
- No stochastic correctors needed

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Conditional Flow Matching

Extension to conditional generation: Instead of unconditional paths, we can define conditional paths $q_t(\mathbf{x} \mid \mathbf{x}_0, c)$ where c represents conditioning information (e.g., class labels, text prompts).

Mathematical formulation: The conditional velocity field becomes:

$$\mathbf{u}_t(\mathbf{x} \mid \mathbf{x}_0, c) := \mathbb{E}\left[\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x}_t \,\middle|\, \mathbf{x}_t = \mathbf{x},\ \mathbf{x}_0,\ c\right] \tag{18}$$

Training objective:

$$J_{\mathsf{CFM}} = \mathbb{E}\left[\, \| \mathbf{v}_\theta(\mathbf{x}_t, t, c) - v \|^2 \,\right] \tag{19}$$

Applications:

- Class-conditional generation
- Text-to-image synthesis
- Image-to-image translation
- Style transfer

Advantage: Maintains the same computational efficiency as unconditional Flow Matching while enabling controlled generation.

Rectified Flow and Optimal Transport

Rectified Flow: A special case of Flow Matching that uses straight-line paths:

$$\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\mathbf{x}_1, \quad v = \mathbf{x}_1 - \mathbf{x}_0 \tag{20}$$

Mathematical properties:

- **Optimal transport**: Minimizes Wasserstein-2 distance
- **Simplicity**: Constant velocity field along each path
- **Efficiency**: Fastest convergence in many cases

Connection to optimal transport: Rectified Flow recovers the optimal transport map between p_0 and p_1 when the data is well-structured.

Practical advantages:

- Fewer function evaluations during sampling
- More stable training
- Better sample quality in many domains

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Flow Matching in Latent Spaces

Motivation: Instead of working directly in pixel space, Flow Matching can be applied in learned latent representations.

Mathematical setup: Given an encoder $E : \mathcal{X} \to \mathcal{Z}$ and decoder $D : \mathcal{Z} \to \mathcal{X}$:

- Learn flow in latent space: $\mathbf{v}_\theta : \mathcal{Z} \times [0, 1] \to \mathcal{Z}$
- Generate latent codes: $\mathbf{z}_0 = \mathbf{z}_1 + \int_0^1 \mathbf{v}_\theta(\mathbf{z}_t, t)dt$
- Decode to images: $\mathbf{x} = D(\mathbf{z}_0)$

Advantages:

- **Computational efficiency**: Lower-dimensional space
- **Sample quality**: Better inductive biases in latent space
- **Flexibility**: Can use different architectures for encoder/decoder

Examples: Latent Diffusion Models (LDMs), Stable Diffusion

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MeanFlow: Motivation

Goal: Achieve single-step (1-NFE) generation while retaining high fidelity and stability.
Problem with standard Flow Matching: Multi-step ODE integration is computationally expensive and can accumulate errors.
Key insight: Learn a *ground-truth average velocity field* that summarizes an entire time interval.
Mathematical formulation: For any pair $r < t$ and point $\mathbf{z}_t$ along the flow path:

$$\overline{\mathbf{u}}(\mathbf{z}_t, r, t) := \frac{1}{t-r} \int_r^t \mathbf{v}(\mathbf{z}_\tau, \tau)\, \mathrm{d}\tau. \tag{21}$$

Properties:

- $\overline{\mathbf{u}}$ is an **underlying field** induced by $\mathbf{v}$ (no networks yet)
- As $r \to t$, $\overline{\mathbf{u}}(\mathbf{z}_t, r, t) \to \mathbf{v}(\mathbf{z}_t, t)$
- Composition over intervals holds: one big step equals two smaller consecutive steps

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Definition: Average Velocity

For any pair $r < t$ and point $\mathbf{z}_t$ along the flow path:

$$\overline{\mathbf{u}}(\mathbf{z}_t, r, t) \;:=\; \frac{1}{t-r}\int_r^t \mathbf{v}(\mathbf{z}_\tau, \tau)\, \mathrm{d}\tau. \tag{22}$$

- $\overline{\mathbf{u}}$ is an **underlying field** induced by $\mathbf{v}$ (no networks yet).
- As $r \to t$, $\;\overline{\mathbf{u}}(\mathbf{z}_t, r, t) \to \mathbf{v}(\mathbf{z}_t, t)$.
- Composition over intervals holds: one big step equals two smaller consecutive steps (additivity of integrals).

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MeanFlow Identity (core relation)

Starting from Eq. (22):

$$(t-r)\,\overline{\mathbf{u}}(\mathbf{z}_t,r,t) \;=\; \int_r^t \mathbf{v}(\mathbf{z}_\tau,\tau)\,\mathrm{d}\tau. \tag{23}$$

Differentiate w.r.t. t (treating r as constant) and use the fundamental theorem of calculus:

$$\boxed{\overline{\mathbf{u}}(\mathbf{z}_t,r,t) \;=\; \mathbf{v}(\mathbf{z}_t,t) \;-\; (t-r)\,\frac{d}{dt}\,\overline{\mathbf{u}}(\mathbf{z}_t,r,t)} \tag{24}$$

where the total derivative expands to

$$\frac{d}{dt}\,\overline{\mathbf{u}}(\mathbf{z}_t,r,t) \;=\; \underbrace{\mathbf{v}(\mathbf{z}_t,t)}_{\frac{d\mathbf{z}_t}{dt}}\cdot\partial_{\mathbf{z}}\overline{\mathbf{u}}(\mathbf{z}_t,r,t) \;+\; \partial_t\overline{\mathbf{u}}(\mathbf{z}_t,r,t). \tag{25}$$

Interpretation: Eq. (24) links instantaneous and average velocities *without* introducing networks.

Mathematical significance: This identity enables learning average velocity fields that can be used for single-step generation.

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Trainable objective

Parameterize a network $\overline{\mathbf{u}}_\theta(\mathbf{z}, r, t)$. Use the MeanFlow identity to form a target (replace $\mathbf{v}$ by sample-conditional $\mathbf{v}_t$):

$$\text{Target:} \quad \overline{\mathbf{u}}_{\text{tgt}} = \mathbf{v}_t - (t-r)\left(\underbrace{\mathbf{v}_t \cdot \partial_{\mathbf{z}} \overline{\mathbf{u}}_\theta}_{\text{JVP}} + \partial_t \overline{\mathbf{u}}_\theta\right). \tag{26}$$

$$\mathcal{L}(\theta) = \mathbb{E} \left\| \overline{\mathbf{u}}_\theta(\mathbf{z}_t, r, t) - \mathrm{sg}(\overline{\mathbf{u}}_{\text{tgt}}) \right\|_2^2. \tag{27}$$

Key components:

- $\mathrm{sg}(\cdot)$: stop-gradient; avoids higher-order differentiation
- JVP (Jacobian-vector product) computed efficiently via autodiff
- If $r = t$, the correction term vanishes $\Rightarrow$ reduces to standard Flow Matching

Mathematical insight: This objective learns to predict average velocities over time intervals, enabling single-step generation.

Pseudocode (training)

MeanFlow training algorithm:

1. Sample $t, r \sim \text{LogitNormal}$; set $t \leftarrow \max(r, t)$, $r \leftarrow \min(r, t)$
2. Sample $\mathbf{x} \sim p_{\text{data}}$, $\epsilon \sim \mathcal{N}(0, I)$; set $\mathbf{z}_t = (1-t)\mathbf{x} + t\epsilon$, $\mathbf{v}_t = \epsilon - \mathbf{x}$
3. Compute $\left(\overline{\mathbf{u}}_\theta,\ \frac{d}{dt}\overline{\mathbf{u}}_\theta\right) = \mathrm{JVP}\left(\overline{\mathbf{u}}_\theta(\cdot, r, t),\ (\mathbf{v}_t, 0, 1)\right)$
4. Set $\overline{\mathbf{u}}_{\text{tgt}} \leftarrow \mathbf{v}_t - (t-r)\,\frac{d}{dt}\overline{\mathbf{u}}_\theta$
5. Minimize $\|\overline{\mathbf{u}}_\theta(\mathbf{z}_t, r, t) - \mathrm{sg}(\overline{\mathbf{u}}_{\text{tgt}})\|_2^2$ (optionally with adaptive weights)

Key innovations:

- **LogitNormal sampling**: Ensures proper time interval distribution
- **JVP computation**: Efficient higher-order derivatives
- **Stop-gradient**: Prevents training instability

Sampling (1 step or few steps)

Replace the time integral by the learned average velocity:

$$\mathbf{z}_r = \mathbf{z}_t - (t - r)\,\overline{\mathbf{u}}_\theta(\mathbf{z}_t, r, t). \tag{28}$$

1-NFE sampling: take $(r, t) = (0, 1)$, $\mathbf{z}_1 = \epsilon$,

$$\boxed{\mathbf{x} \equiv \mathbf{z}_0 = \epsilon - \overline{\mathbf{u}}_\theta(\epsilon, 0, 1)}\,. \tag{29}$$

Mathematical significance: This single-step generation maintains the same theoretical guarantees as multi-step Flow Matching.

Practical advantages:

- **Speed**: Single function evaluation
- **Stability**: No accumulation of integration errors
- **Flexibility**: Can still use few-step variants by chaining intervals

CFG inside the field (still 1-NFE)

Define a CFG velocity field

$$\mathbf{v}_{\text{cfg}}(\mathbf{z}_t, t \mid c) \;=\; \omega\, \mathbf{v}(\mathbf{z}_t, t \mid c) + (1-\omega)\, \mathbf{v}(\mathbf{z}_t, t). \tag{30}$$

Key insight: CFG can be applied directly to the average velocity field:

$$\overline{\mathbf{u}}_{\text{cfg}}(\mathbf{z}_t, r, t \mid c) \;=\; \omega\, \overline{\mathbf{u}}(\mathbf{z}_t, r, t \mid c) + (1-\omega)\, \overline{\mathbf{u}}(\mathbf{z}_t, r, t). \tag{31}$$

Mathematical property: This maintains the 1-NFE property while enabling conditional generation.
Practical significance: MeanFlow can achieve both single-step generation and strong conditioning, combining the best of both worlds.

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Key Takeaways: Flow Matching

Mathematical foundations:

- Flow Matching learns velocity fields that transport probability distributions via the continuity equation
- Conditional paths provide analytic supervision without score estimation
- The method connects to optimal transport theory and diffusion models

Computational advantages:

- No score estimation during training
- Deterministic sampling with flexible step sizes
- Stable gradients and efficient training

Practical benefits:

- Fewer function evaluations than diffusion models
- Better sample quality in many domains
- Natural extension to conditional generation

MeanFlow extension: Enables single-step generation while maintaining theoretical guarantees.

Future Directions and Open Problems

Theoretical developments:

- **Path optimization**: Finding optimal conditional paths for specific data distributions
- **Convergence analysis**: Understanding when Flow Matching achieves optimal transport
- **Error bounds**: Quantifying approximation errors in finite-sample settings

Practical improvements:

- **Architecture design**: Specialized networks for different data modalities
- **Training strategies**: Adaptive loss weighting and curriculum learning
- **Sampling efficiency**: Better ODE solvers and step size selection

Applications:

- **Scientific computing**: Molecular dynamics, fluid simulation
- **Creative AI**: Music, video, and 3D content generation
- **Data augmentation**: Synthetic data for training other models

Outline

北京雁栖湖
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Key References

Core Flow Matching papers:

- Lipman et al., *Flow Matching: A unifying perspective of score-based training and generative modeling* (2023)
- Liu et al., *Flow Matching for Generative Modeling* (2023)
- Albergo et al., *Building Normalizing Flows with Stochastic Interpolants* (2023)

MeanFlow and extensions:

- Song et al., *MeanFlow: A Flow Matching Method for One-Step Generation* (2024)
- Song et al., *Rectified Flow: A Marginal Preserving Approach to Optimal Transport* (2022)

Related work:

- Song et al., *Score-based generative modeling through stochastic differential equations* (2021)
- Ho et al., *Denoising Diffusion Probabilistic Models* (2020)
- Chen et al., *Neural Ordinary Differential Equations* (2018)

Implementation resources:

- Flow Matching Guide and Code (GitHub)
- Open source libraries: diffusers, torchdyn

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Welcome inputs

Any comments?

Welcome your inputs!

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